

**Tilmovet® 200 g/kg
Premix for Medicated Feeding Stuff
for Pigs and Rabbits**

Keep out of reach of children.

STATEMENT OF ACTIVE SUBSTANCE:

200 g tilmicosin per kg

PRODUCT DESCRIPTION:

Tilmovet® 200 g/kg Premix for Medicated Feeding Stuff for Pigs and Rabbits represents yellowish to brown granules.

PHARMACODYNAMICS:

Tilmicosin is a mainly bactericidal semi-synthetic antibiotic of the macrolide group. It is believed to affect the bacterial protein synthesis *in vitro* and *in vivo*, without affecting the nucleic acid synthesis. It is mostly bacteriostatic. It has a bactericidal effect on *Pasteurella* spp.

Tilmicosin has a wide spectrum of activity against Gram-positive organisms is particularly active against *Pasteurella*, *Actinobacillus* (*Haemophilus*) and *Mycoplasma* organisms of bovine, porcine and avian origin Tilmicosin has some activity against certain Gram-negative micro-organisms.

Cross resistance between tilmicosin and other macrolide antibiotics has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase.

Ribosomal methylase, encoded by the *erm* gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site.

The gene that encodes for an efflux mechanism, *mef*, also brings about a moderate degree of resistance.

Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as *acrAB* genes. Resistance of *Pseudomonas* species and other Gram negative bacteria, enterococci and staphylococci may be precipitated by chromosomally controlled alteration of permeability or uptake of the drug.

PHARMACOKINETICS:

Pigs:

Absorption: When administered to pigs via the oral route at a dose of 400 mg tilmicosin/kg feed (equivalent to approximately 21.3 mg tilmicosin/kg bodyweight/day), tilmicosin moves rapidly out of the serum into areas of low pH. The highest concentration in the serum (0.23 ±0.08 µg/ml) was recorded on day 10 of medication, but concentrations above the limit of quantification (0.10 µg/ml) were not found in 3 out of 20 animals examined. Lung concentrations increased rapidly between days 2 and 4 but no significant changes were obtained following four days of dosing. The maximum concentration in lung tissue (2.59±1.01 µg/ml) was recorded on day 10 of medication.

When administered at a dose of 200 mg tilmicosin/kg feed (equivalent to approximately 11.0 mg/kg/day), plasma concentrations above the limit of quantification (0.10 µg/ml) were found in 3 out of 20 animals examined. Quantifiable levels of tilmicosin were found in lung tissue with the maximum concentration (1.43±1.13 µg/ml) being recorded on day 10 of medication.

Distribution: Following oral administration, tilmicosin is distributed throughout the body with especially high levels found in the lung and in lung tissue macrophages. It is also distributed in the liver and kidney tissues.

Rabbits:

Absorption: When administered orally to rabbits at a dose of 12 mg tilmicosin/kg b.w. as a single dose there is a quick absorption. Maximum concentrations were reached in 30 minutes, being the Cmax obtained of 0.35 µg/ml. Tilmicosin plasma concentrations decreased to 0.1 µg/ml within 2 hours and to 0.02 µg/ml after 8 hours. The elimination half-life was 22 hours.

Distribution: Following oral administration, tilmicosin is distributed throughout the body with especially high levels found in lungs. After 5 days of treatment with medicated feed at a dosage of 200 ppm of product, tilmicosin concentrations in lung tissues were of 192 ± 103 µg/g.

Applicable to both species:

Biotransformation: Several metabolites are formed, the predominant one being identified as T1. However the bulk of tilmicosin is excreted unchanged.

Elimination: Following oral administration, tilmicosin is excreted mainly via the bile into the faeces but a small proportion is excreted via the urine.

INDICATIONS FOR USE:

Pigs: Prevention and treatment of respiratory disease caused by *Actinobacillus pleuropneumoniae*, *Mycoplasma hyopneumoniae*, *Pasteurella multocida* and other organisms sensitive to tilmicosin

Rabbits: Prevention and treatment of respiratory disease caused by *Pasteurella multocida* and *Bordetella bronchiseptica*, susceptible to tilmicosin.

RECOMMENDED DOSAGE:

Pigs

Administer in the feed at a dose of 8 to 16 mg/kg body weight/day of tilmicosin (equivalent to 200 to 400 ppm in the feed) for a period of 15 to 21 days. *

Rabbits

Administer in the feed at 12 mg/kg body weight/day of tilmicosin (equivalent to 200 ppm in the feed) for 7 days. **

MODE OF ADMINISTRATION:

The uptake of medicated feed depends on the clinical condition of the animals. In order to obtain a correct dosage the concentration of tilmicosin has to be adjusted accordingly. Use the following formula: ***

To ensure thorough dispersion of the product, it should first be mixed with a suitable quantity of feed before incorporation into the finished feed.

This product can be incorporated into

pelleted feed, preconditioned for the minimum time-period at a temperature not exceeding 75°C.

CONTRAINDICATIONS:

Horses or other *Equidae*, must not be allowed access to feeds containing tilmicosin. Horses fed with tilmicosin medicated feeds may present signs of toxicity with lethargy, anorexia, reduction of feed consumption, loose stools, colic, distension of the abdomen and death.

Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.

Do not use in animals hypersensitive to tilmicosin and when there is resistance to tilmicosin or cross resistance to other macrolides like tylosin, erythromycin or lincomycin

WARNINGS AND PRECAUTIONS:

Special warnings (for each target species)

With regard to the management of respiratory disease outbreaks, it should be noted that acutely ill animals are likely to be inappetent and therefore require parenteral treatment.

Special precautions for use in animals

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substances.

Due to the likely variability (time, geographical) in the occurrence of the resistance of bacteria for tilmicosin, bacteriological sampling and susceptibility testing are recommended.

Cross-resistance between tilmicosin and other macrolide antibiotic has been observed. Use of the product should be based on susceptibility testing and take into account official, national and regional antimicrobial policies. Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substances.

Special precautions for the person administering the veterinary medicinal product to animals

The handling of the product in case of known hypersensitivity to macrolide antibiotics must be avoided.

May cause sensitisation by skin contact. May cause skin and eye irritation. Avoid direct skin contact. Wear overalls, safety glasses and impervious gloves when mixing and handling the product. Wash affected parts if skin contact occurs. If accidental eye contact occurs, immediately rinse thoroughly with water. In case of accidental ingestion, or if you develop symptoms following exposure such as skin rash, seek medical advice immediately and show the package leaflet or the label to the physician. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

If the operations involve the risk of exposure to dust, wear either a disposable filter and half mask respirator. This warning is particularly relevant to on-farm mixing, where the risk of exposure to dust is likely to

be enhanced.

INTERACTIONS WITH OTHER MEDICAMENTS:

Do not use simultaneously with other macrolides and lincosamides.

Do not use simultaneously with bacteriostatic antimicrobial agents.

Tilmicosin may less the antibacterial activity of β-lactam antibiotics.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION:

Laboratory studies in rats have not produced any evidence of a teratogenic, foetotoxic/embryotoxic effect of tilmicosin, however, a maternotoxicity was observed at doses that were close to the therapeutic dosage. The product is safe in sows whatever the pregnancy stages.

The safety of the veterinary medicinal product has not been established in boars used for breeding purposes.

ADVERSE EFFECTS/UNDESIRABLE EFFECTS:

In very rare cases (less than 1 animal in 10,000 animals), feed intake may decrease (including feed refusal) in animals receiving medicated feed. This effect is transient.

OVERDOSE AND TREATMENT:

No symptoms of overdose have been seen in pigs fed a ration containing levels of tilmicosin up to 80 mg/kg bodyweight (equivalent to 2000 ppm in the feed or ten times the recommended dose) for 15 days.

INCOMPATIBILITIES:

Do not mix into feed containing bentonite. In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other veterinary medicinal products.

WITHDRAWAL PERIODS:

Pigs: meat and offal: 21 days

Rabbits: meat and offal: 4 days

DISPOSAL

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

SHELF-LIFE OF THE VETERINARY PRODUCT AS PACKAGED FOR SALE:

3 years.

SHELF LIFE AFTER FIRST OPENING OF THE CONTAINER: 3 months.

SHELF LIFE AFTER INCORPORATION INTO MEAL OR PELLETED FEED:

3 months.

STORAGE CONDITIONS:

Do not store above 30°C.

Store in the original container.

Store in a dry place.

PACKAGING AVAILABLE:

20 kg paper/polyethylene bag.

1 kg resealable block bottom zipped sachet made of polyethylene / aluminium/ polyethylene terephthalate

MANUFACTURER:

BIOVET Joint Stock Company
39 Petar Rakov Street, 4550 Peshtera, Bulgaria

MARKETING AUTHORIZATION

HOLDER:

Pahang Pharmacy Sdn. Bhd. (10826-K)
Lot 5979 Jalan Teratai, 5½ Mile Off Jalan Meru, 41050 Klang, Selangor Darul Ehsan, Malaysia

DATE OF REVISION:

October, 2019.

* Indication	Dose of tilmicosin	Duration of treatment	Inclusion rate in feed
Prevention and treatment of respiratory disease	8-16 mg/kg bodyweight/day	15 to 21 days	1-2 kg product /tonne

** Indication	Dose of tilmicosin	Duration of treatment	Inclusion rate in feed
Prevention and treatment of respiratory disease	12 mg/kg bodyweight/day	7 days	1 kg product /tonne

$$*** \text{ Kg product/tonne feed} = \frac{\text{Dose rate (mg/kg bodyweight)} \times \text{bodyweight (kg)}}{\text{Daily feed intake (kg) x premix strength (g/kg)}}$$